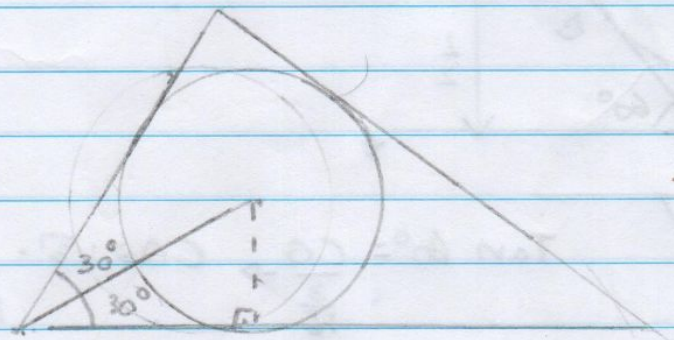
Beatriz Bastos Borges Nº 03 CTII 350

Tarefa Básica

1.

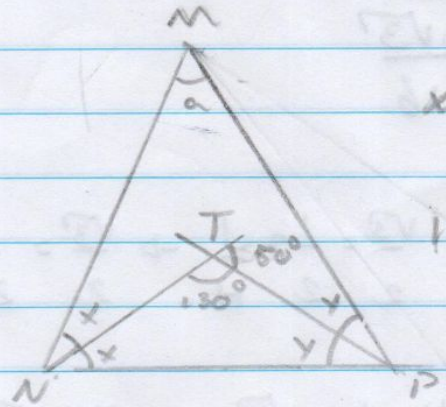


$\text{sen } 30° = \frac{1}{H}$

$\frac{1}{2} = \frac{1}{H} \rightarrow \boxed{H = 2}$

$\boxed{\text{Alternativa D}}$

2.



$x + y = 180° - 130° \rightarrow x + y = 50°$

$130° = 50° + a \rightarrow \boxed{a = 80°}$

$\boxed{\text{Alternativa E}}$

3. A) Pode não ser possível caso os pontos A e B estejam perto do C

B) Não há informação o suficiente para afirmar.

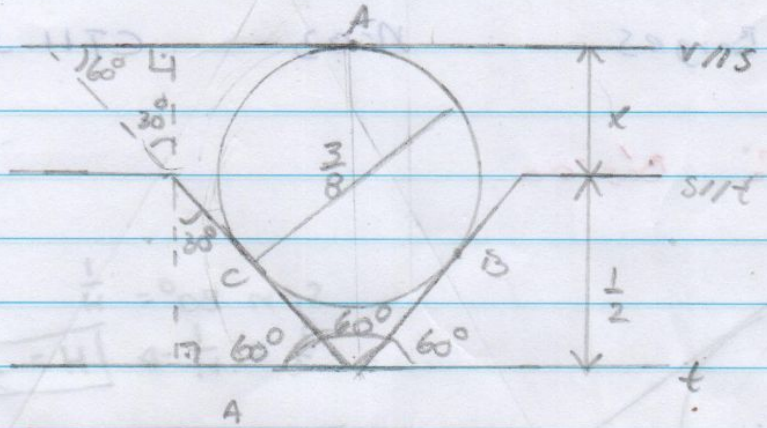
C) O mesmo motiva de A

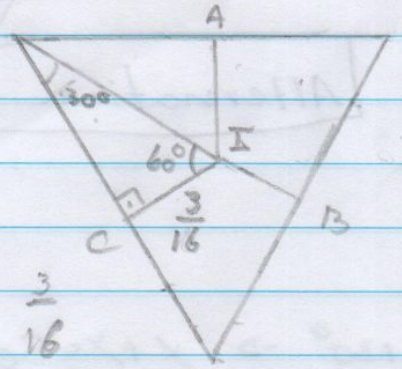
D) É possível caso os pontos A e B estejam a mesma distancia de C, tendo assim a possibilidade de ter uma reta passando por esses dois pontos.

E) Unica que sobrou $\boxed{\text{Alternativa E}}$.

SÃO DOMINGOS S.A.

4.



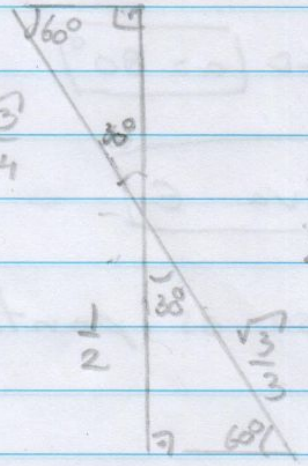


$\frac{3}{8} : 2$

$\frac{3}{8} \cdot \frac{1}{2} = \frac{3}{16}$

$\text{Tan } 60° = \frac{CO}{\frac{3}{16}} \rightarrow CO = \sqrt{3} \cdot \frac{3}{16}$

$CO = \frac{3\sqrt{3}}{16}$



$\text{Sen } 60° = \frac{\frac{1}{2}}{H} \rightarrow \frac{\sqrt{3}}{2} = \frac{1}{2} \cdot \frac{1}{H} \rightarrow \frac{\sqrt{3}}{2} = \frac{1}{2H}$

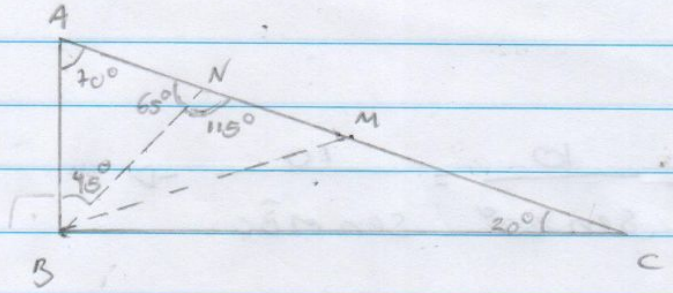
$2H\sqrt{3} = 2 \rightarrow H = \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} \rightarrow H = \frac{\sqrt{3}}{3}$

$2 \cdot \frac{3\sqrt{3}}{16} \rightarrow \frac{3\sqrt{3}}{8}$ $\left\{ \frac{3\sqrt{3}}{8} - \frac{\sqrt{3}}{3} \rightarrow \frac{9\sqrt{3} - 8\sqrt{3}}{24} \rightarrow \frac{\sqrt{3}}{24} \right.$

$\text{Sen } 60° = \frac{CO}{\frac{\sqrt{3}}{24}} \rightarrow CO = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{24} \rightarrow CO = \frac{3 : 3}{48 : 3} \rightarrow \boxed{CO = \frac{1}{16}}$

$\boxed{\text{Alternativa E}}$

5.



$\overline{AB} = 20\ cm$

$\boxed{\overline{AM} = \overline{MC} = 10\ cm}$

$\cos 70° = \dfrac{AB}{20} \rightarrow AB = \cos 70° \cdot 20$

$AB^2 = AM^2 + BM^2 - 2 \cdot AM \cdot BM \cdot \cos A\hat{M}B$

$(\cos 70° \cdot 20)^2 = 10^2 + BM^2 - 2 \cdot 10 \cdot BM \cdot \cos A\hat{M}B$

$(\cos 70°) \cdot 400 = 100 + BM^2 - 20 \cdot BM \cdot \cos A\hat{M}B$ (I)

$\operatorname{sen} 70° = \dfrac{BC}{20} \rightarrow BC = \operatorname{sen} 70° \cdot 20$

$BC^2 = MC^2 + BM^2 - 2 \cdot MC \cdot BM \cdot \cos B\hat{M}C$

$(20 \cdot \operatorname{sen} 70°)^2 = 10^2 + BM^2 - 2 \cdot 10 \cdot BM \cdot (-\cos A\hat{M}B)$

$(\operatorname{sen} 70°)^2 \cdot 400 = 100 + BM^2 + 20 \cdot BM \cdot \cos A\hat{M}B$ (II)

I + II

$(\cos 70°)^2 \cdot 400 + (\operatorname{sen} 70°)^2 \cdot 400 = 100 + BM^2 - 20 \cdot BM \cdot \cos A\hat{M}B +$
$100 + BM^2 + 20 \cdot BM \cdot \cos A\hat{M}B \rightarrow$

$400\,((\cos 70°)^2 + (\operatorname{sen} 70°)^2) = 200 + 2 \cdot BM^2 \rightarrow 400 = 200 + 2 \cdot BM^2$

$200 = 2 \cdot BM^2 \rightarrow BM^2 = \sqrt{100} \rightarrow BM = 10\ cm$

SÃO DOMINGOS S.A.

$A\hat{B}N = N\hat{B}C = 45^\circ$

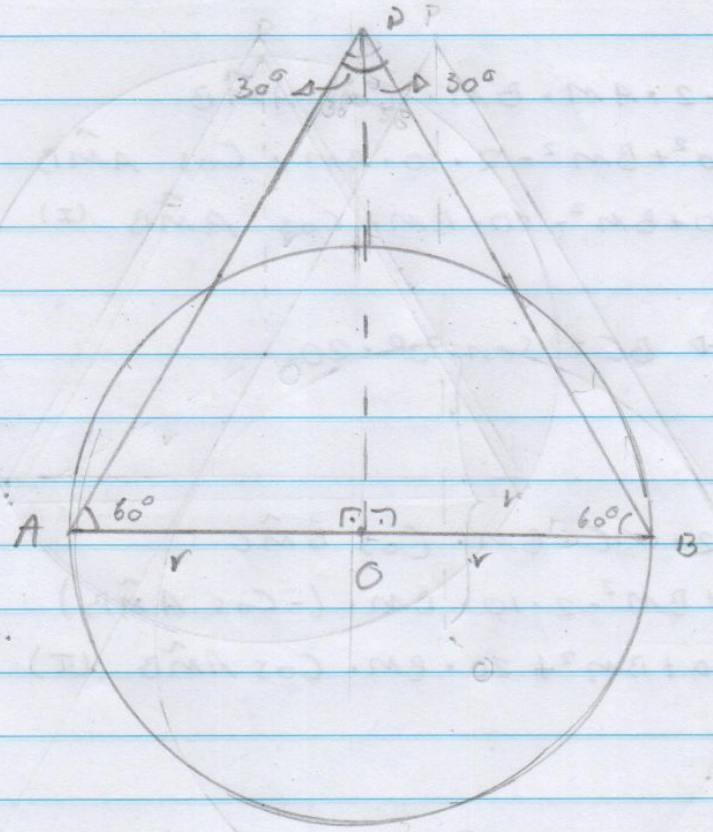
$$\frac{BM}{\text{sen } 20^\circ} = \frac{MC}{\text{sen } M\hat{B}C} \rightarrow \frac{10}{\text{sen } 20^\circ} = \frac{10}{\text{sen } M\hat{B}C} \rightarrow$$

$10 \cdot \text{sen } M\hat{B}C = 10 \cdot \text{sen } 20^\circ \rightarrow \text{sen } M\hat{B}C = \text{sen } 20^\circ$

$\therefore M\hat{B}C = 20^\circ$

$45^\circ = M\hat{B}N + 20^\circ \rightarrow \boxed{M\hat{B}N = 25^\circ}$

6.



$$\text{sen } 30^\circ = \frac{r}{PO} \rightarrow \frac{1}{2} = \frac{r}{PO} \rightarrow \boxed{PO = 2r}$$

$\boxed{\text{Alternativa C}}$